

RESEARCH

Open Access



Option pricing mechanisms driven by backward stochastic differential equations

Yufeng Shi^{1,2,3*} , Bin Teng¹ and Sicong Wang^{1*}

*Correspondence:
yfshi@sdu.edu.cn;
wangsicong@mail.sdu.edu.cn

¹ Institute for Financial Studies
and School of Mathematics,
Shandong University, Jinan,
China

² National Center for Applied
Mathematics of Shandong,
Shandong University, Jinan,
China

³ State Key Laboratory
of Cryptography and Digital
Economy Security, Shandong
University, Jinan, China

Abstract

This study investigates an option pricing method called g -pricing based on backward stochastic differential equations combined with deep learning. We adopted a data-driven approach to find a market-appropriate generator of the backward stochastic differential equation, which is achieved by leveraging the universal approximation capabilities of neural networks. Option pricing, which is the solution to the equation, is approximated using a recursive procedure. The empirical results for the S&P 500 index options show that the proposed deep learning g -pricing model has lower absolute errors than the classical Black–Scholes–Merton model for the same forward stochastic differential equations. The g -pricing mechanism has potential applications in option pricing.

Keywords: Option pricing, Backward stochastic differential equation, Numerical method, Deep learning

Introduction

The Black–Scholes–Merton (BSM) model is the foundation of the derivative pricing theory (Black and Scholes 1973, Merton 1973). Commonly used derivative pricing models use the BSM model as the basis for modeling risky assets using stochastic differential equations (SDEs). There are different assumptions about the volatility of the underlying asset, with the most representative being the local and stochastic volatility models. The local volatility model assumes that volatility is a deterministic function of time t and the underlying asset price X_t , such as the constant elastic variance (CEV) model (Cox 1997) and volatility fitting methods (Dupire 1994, Derman and Kani 1994). Stochastic volatility models introduce volatility processes such as the Hull–White model (Hull and White 1987) and the Heston model (Heston 1993). In addition, local stochastic volatility models represented by the stochastic alpha, beta, rho (SABR) model (Hagan et al. 2002) combine the advantages of local and stochastic volatility models. These volatility models typically provide explicit or semi-explicit pricing formulas. However, owing to the uncertainty in the financial market, these models are usually subject to varying degrees of model bias and require frequent recalibration based on market data. For example, Eraker et al. (2003) point out that the risk of volatility change is underestimated in the above models. For further advances in option pricing and volatility modeling, readers

are referred to the following: Madan et al. (1998), Cont and Da Fonseca (2002), Gatheral (2004), Gatheral and Jacquier (2014), Cohen et al. (2023), Guyon and Lekeufack (2023).

However, El Karoui et al. (1997) found that the theoretical framework of backward stochastic differential equations (BSDEs) established by Pardoux and Peng (1990) is applicable to pricing derivative securities such as futures and options. The BSM partial differential equation (PDE) is related to a specific linear BSDE using the nonlinear Feynman–Kac formula theorem (Peng 1991, Pardoux and Peng 1992). Thus, the BSDE theory is a more fundamental and general computational tool for the quantitative analysis of derivative securities pricing. On this basis, Peng (1997) defined g -expectations using the solutions of a specific class of BSDEs. Chen and Epstein (2002) applied the g -expectation theory to obtain the Chen–Epstein asset pricing formula. Furthermore, Gianin (2006) showed that the g -expectation satisfies the axiom of the consistent risk measure proposed by Artzner et al. (1999). Thus, the g -expectation is a time-consistent dynamic nonlinear expectation. Peng (2012) discusses a derivative pricing mechanism based on the BSDE generator g . Under the g -pricing framework, the price of an option is represented by the generator function g of the BSDE. Thus, Peng (2012) converted the option pricing problem into finding the generator g based on actual data from the options market.

There have been a few preliminary attempts involving methods such as test methods, nonparametric statistical approaches, and g -submartingale decomposition techniques to identify the generator g as in Yang (2006) and Chen (2007). However, in complex real-world scenarios, these methods tend to be oversimplified, making them less effective in dealing with the inherent complexity of reality. By contrast, modern data science techniques offer a more robust, data-driven model selection mechanism. Every given model only approximates realistic data, and the risks associated with inappropriate models are challenging to detect and quantify. Deep learning provides a more effective data-driven model-selection mechanism. Deep learning methods have been applied to address the inverse problems of PDEs, which involve inferring unknown PDE structures from data. The physics-informed neural network model proposed by Raissi et al. (2019) utilizes deep neural networks to estimate a physical model from limited data, transforming the process of solving PDEs into an optimization problem for the network parameters. For more approaches based on deep learning for constructing or inferring (unknown) PDEs from existing data, one may refer to Long et al. (2019) and Li et al. (2020). The deep BSDE method proposed by E et al. (2017) and Han et al. (2018) numerically solves forward–backward stochastic differential equations (FBSDEs) and the associated parabolic PDEs by introducing deep neural networks. Building on this, Ji et al. (2020) introduced neural networks as control terms to solve FBSDEs. By leveraging these numerical methods, we combined deep learning with BSDEs to find derivative prices and the corresponding hedging strategies by incorporating real market data. The integration of deep neural networks simplifies the implementation of finding the generator g and allows it to be determined simultaneously with the numerical solutions of the BSDEs. From a data-driven perspective, this approach enables us to obtain a generator g that is more appropriate for the market.

In this study, we consider an option pricing method driven by the BSDE in terms of the g -pricing mechanism. We suggest that option prices depend on two factors: (1) the estimation of the volatility of the underlying asset, which is in response to the forward

SDE parameter estimation, and (2) the property of the nonlinear option instruments and market conditions (e.g., transaction costs and interest rates), which correspond to the estimation of the generator g of BSDE. Under these assumptions, we introduced a deep learning-based g -pricing model that incorporates a numerical solution method for BSDEs, extracting the implicit expression of generator g from historical prices in the real options market. Additionally, we conducted an empirical analysis of S&P 500 index (SPX) options.

The remainder of this study is organized as follows. Section 2 briefly describes the theoretical properties of the g -pricing mechanism in Peng (2012) and its relationship with BSM model. We propose the deep learning g -pricing method in Sect. 3. Section 4 contains the structures and parameters of the deep network, and analyzes the empirical results of the deep learning g -pricing method on SPX European options. In Sect. 5, we discuss the modeling methodology and the empirical results of the g -pricing method. Finally, Sect. 6 concludes the paper.

The g -pricing mechanism driven by BSDEs

In this section, we focused on the g -pricing mechanism driven by BSDEs, as introduced in Peng (2012) and explored some of its properties in the context of pricing.

The g -pricing framework

Suppose that the underlying asset price process $\{X_t\}_{t \geq 0}$ is driven by the Brownian motion $\{W_t\}_{t \geq 0}$ in the probability space $(\Omega, \mathcal{F}, \mathbb{P})$. Let $\mathcal{N}$ denote the class of $\mathbb{P}$ -null elements of $\mathcal{F}$. For each $t \in [0, T]$, we defined the filtration generated by Brownian motion $\mathcal{F}_t^W = \sigma\{W_r - W_0; 0 \leq r \leq t\} \vee \mathcal{N}$.

Consider the European claim Y_T settled at time T , which corresponds to the underlying asset price X_T . The risk-free rate is r . We denote the European option price process as $\{Y_t\}_{0 \leq t \leq T}$. For any t , X_t and Y_t are $\mathcal{F}_t^W$ -measurable square integrable random variables, denoted as $X_t \in L^2(\mathcal{F}_t^W)$ and $Y_t \in L^2(\mathcal{F}_t^W)$. For simplicity, we considered the case in which both X_t and Y_t are both 1-dimensional.

Suppose that the joint dynamics of the underlying asset price X_t and associated option price Y_t are described by the following FBSDEs:

$$\begin{cases} X_t = x_0 + \int_0^t b(s, X_s) ds + \int_0^t \sigma(s, X_s) dW_s, & \text{(SDE)} \\ Y_t = Y_T + \int_t^T g(s, X_s, Y_s, Z_s) ds - \int_t^T Z_s dW_s. & \text{(BSDE)} \end{cases} \quad (1a) \quad (1b)$$

The forward SDE (1a) describes the underlying stock price process starting from x_0 at time zero. Function $\sigma : (t, x) \in [0, +\infty) \times \mathbb{R} \rightarrow \mathbb{R}$ represents the local volatility of the stock. Generator $g : (t, x, y, z) \in [0, +\infty) \times \mathbb{R} \times \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ represents the pricing mechanism. If terminal Y_T is the payoff of a European call/put option, then the solution Y_0 of BSDE (1b) describes the option price.

With a given generator g and various terminal times $T > t$ along with the corresponding terminal conditions Y_T , the BSDEs (1b) characterize the prices of different option contracts. For example, in the case of a European call option with expiration date T and

strike price K , the terminal condition is $Y_T = (X_T - K)^+$, whereas the put option corresponds to the terminal condition $Y_T = (K - X_T)^+$.

According to Pardoux and Peng (1990), when the generator g (as well as b and σ) satisfies the Lipschitz continuity condition and the terminal Y_T satisfies $Y_T \in L^2(\mathcal{F}_T^W)$, the BSDE has a unique square-integrable $\mathcal{F}_t^W$ -adapted (measurable) solution $(Y_t, Z_t)_{t \in [0, T]}$. Pardoux and Peng (1992) established the connection between the solutions of a type of parabolic PDEs and BSDEs, called the nonlinear Feynman–Kac formula. For an SDE starting from (t, x) , the solution of the BSDE (1b) can be expressed as $Y_t^{t,x} = u(t, x)$ and $Z_t^{t,x} = \partial_x u \sigma(t, x)$, where $u(t, x)$ is a deterministic function satisfying the following PDE:

$$\begin{cases} u(t, x) = \phi(x) + \int_t^T [\mathcal{L}u(s, x) + g(s, x, u(s, x), \partial_x u \sigma(s, x))] ds, \\ u(T, x) = \phi(x), \end{cases} \quad (2)$$

where $\mathcal{L}$ is a second-order differential operator defined as follows:

$$\mathcal{L}u(t, x) = \frac{1}{2} \sigma^2(t, x) \frac{\partial^2}{\partial x^2} u(t, x) + b(t, x) \frac{\partial}{\partial x} u(t, x).$$

In BSDE theory, the term Z serves as the control variable. According to PDE (2), $Z_t^{t,x} = \partial_x u \sigma(t, x)$. Here, $\partial_x u$ denotes the partial derivative of $u(t, x)$ with respect to x representing the rate of change in the option price relative to changes in the underlying asset price, which corresponds to the Greek Delta. Consequently, the control term Z contains information relevant to the hedging strategy. When both Z and σ are estimated, the corresponding hedging strategy can be effectively determined.

The example of BSM formula and its related BSDE

In this section, we explained the application of BSDEs to European options pricing using the BSM formula as an example. Given stock price x_0 , suppose that the risk-free rate is r and that volatility is a constant σ . The BSM prices for European call and put options with strike price K and maturity date T are given by:

$$Y_0^c = x_0 N(d_1) - Ke^{-rT} N(d_2), \quad (3)$$

$$Y_0^p = Ke^{-rT} N(-d_2) - x_0 N(-d_1), \quad (4)$$

respectively, where

$$d_1 = \frac{\log \frac{x_0}{K} + \left(r + \frac{\sigma^2}{2}\right) T}{\sigma \sqrt{T}},$$

$$d_2 = \frac{\log \frac{x_0}{K} + \left(r - \frac{\sigma^2}{2}\right) T}{\sigma \sqrt{T}},$$

where $N(\cdot)$ denotes the cumulative distribution function of the standard normal distribution.

El Karoui et al. (1997) and Peng (2012) demonstrate that Eqs. (3) and (4) are solutions of the following (linear) FBSDEs with specific terminal conditions:

$$\begin{cases} X_t = x_0 + \int_0^t bX_s ds + \int_0^t \sigma X_s dW_s, \\ Y_t = \phi(X_T) + \int_t^T \left(-rY_s - \frac{b-r}{\sigma} Z_s \right) ds - \int_t^T Z_s dW_s. \end{cases} \quad (5a)$$

$$(5b)$$

When the terminal condition is $\phi(X_T) = (X_T - K)^+$, the solution of Eq. (5b) is the call option price Y_0^c in Eq. (3). Similarly, the solution is Y_0^p if the terminal condition $\phi(X_T)$ is consistent with the put option's payoff $(K - X_T)^+$. Furthermore, the solutions Z_0 in Eq. (5b) are $Z_0^c = N(d_1) \sigma x_0$ and $Z_0^p = -N(-d_1) \sigma x_0$ respectively. In the BSM framework, $N(d_1)$ is the delta of the European call option, which represents the option positions in the dynamic hedge. This suggests that process Z in BSDE (5b) contains information about the hedging positions.

Additionally, the nonlinear Feynman–Kac formula given by Pardoux and Peng (1992) establishes a connection between BSM PDE and the linear FBSDEs (5a)–(5b). When the terminal condition $Y_T = \phi(X_T)$ is given, the forward SDE (5a) starts from $x \in \mathbb{R}$ at time t , and the solution $Y_t^{t,x} = u(t, x)$ of BSDE (5b) satisfies the following partial differential equation:

$$\begin{cases} ru(t, x) = \frac{\partial u}{\partial t}(t, x) + \frac{1}{2} \sigma^2 x^2 \frac{\partial^2 u}{\partial x^2}(t, x) + rx \frac{\partial u}{\partial x}(t, x), \\ u(T, x) = \phi(x). \end{cases} \quad (6a)$$

$$(6b)$$

Consequently, FBSDEs can be regarded as probabilistic interpretations of PDE systems. This forms the theoretical foundation for modeling option-pricing problems using FBSDEs.

Remark 1 It is necessary to note that the BSM PDE (6a) does not contain the expected returns b of the stock price process (5a). When the terminal function $\phi(x)$ is given, the solution of PDE (6a) is the same as that of the forward–backward stochastic system (5a)–(5b), as discussed in Pardoux and Peng (1992), and both are independent of b .

The conditions and properties of the g -pricing mechanism

For a given terminal time $0 < T < \infty$, the solutions of the FBSDEs (1a)–(1b) are $Y_t = \mathcal{E}_{t,T}^g[Y_T]$ and $t \in [0, T]$. Subsequently, $\mathcal{E}_{t,T}^g[\cdot] : L^2(\mathcal{F}_T^W) \rightarrow L^2(\mathcal{F}_t^W)$ is the g -pricing mechanism in Peng (2012). Proposition 1 provides the theoretical properties of the g -pricing mechanism. Without further explanation, the equations and inequalities involving random variables almost certainly hold true.

Proposition 1 (Peng (2012)) *Consider the option-pricing mechanism $\mathcal{E}_{t,T}^g[\cdot]$ driven by the FBSDEs (1a)–(1b). Let the generator $g : (t, x, y, z) \in [0, +\infty) \times \mathbb{R} \times \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ satisfy the Lipschitz continuity conditions. In other words, there exists a constant $K > 0$ such that for $\forall t > 0, x, x' \in \mathbb{R}, y, y' \in \mathbb{R}, z, z' \in \mathbb{R}$,*

$$|g(t, x, y, z) - g(t, x', y', z')| \leq K(|x - x'| + |y - y'| + |z - z'|).$$

Then, the g -pricing mechanism is $\mathcal{F}_t^W$ -consistent; that is, for each $0 \leq r \leq s \leq t \leq T$ and $Y_t, Y_t' \in L^2(\mathcal{F}_t^W)$, it satisfies

- (A1) $\mathcal{E}_{s,t}^g[Y_t] \geq \mathcal{E}_{s,t}^g[Y'_t]$, if $Y_t \geq Y'_t$;
- (A2) $\mathcal{E}_{t,t}^g[Y_t] = Y_t$;
- (A3) $\mathcal{E}_{r,s}^g[\mathcal{E}_{s,t}^g[Y_t]] = \mathcal{E}_{r,t}^g[Y_t]$;
- (A4) $1_A \mathcal{E}_{s,t}^g[Y_t] = 1_A \mathcal{E}_{s,t}^g[1_A Y_t]$, for any $A \in \mathcal{F}_t^W$.

Proposition 1 provides the most fundamental properties of the pricing mechanism driven by BSDEs. In addition, if further constraints are imposed on generator g , the g -pricing mechanism has the following properties.

- (i) (*self-financing*) If $\forall t > 0, x \in \mathbb{R}, g(t, x, 0, 0) = 0$, then $\mathcal{E}_{s,t}^g[0] = 0, 0 \leq s \leq t$.
- (ii) (*convexity*) If $\forall \alpha \in [0, 1], \forall t > 0, x \in \mathbb{R}, y, y' \in \mathbb{R}, z, z' \in \mathbb{R}, g(t, x, \alpha y + (1 - \alpha)y', \alpha z + (1 - \alpha)z') \leq \alpha g(t, x, y, z) + (1 - \alpha)g(t, x, y', z')$, then $\forall 0 \leq s \leq t \leq T$ and $\forall \xi, \zeta \in \mathcal{F}_t^W, \mathcal{E}_{s,t}^g[\alpha \xi + (1 - \alpha)\zeta] \leq \alpha \mathcal{E}_{s,t}^g[\xi] + (1 - \alpha)\mathcal{E}_{s,t}^g[\zeta]$.
- (iii) (*positively homogeneous*) If $\forall \lambda \geq 0, t > 0, x \in \mathbb{R}, y \in \mathbb{R}, z \in \mathbb{R}, g(t, x, \lambda y, \lambda z) = \lambda g(t, x, y, z)$, then for any $0 \leq s \leq t$ and $\forall \xi \in \mathcal{F}_t^W, \mathcal{E}_{s,t}^g[\lambda \xi] = \lambda \mathcal{E}_{s,t}^g[\xi]$.
- (iv) (*subadditive*) If $\forall t > 0, \forall (y, z), (y', z') \in \mathbb{R} \times \mathbb{R}, g(t, y + y', z + z') \leq g(t, y, z) + g(t, y', z')$, then for any $0 \leq s \leq t$ and $\forall \xi, \zeta \in \mathcal{F}_t^W, \mathcal{E}_{s,t}^g[\xi + \zeta] \leq \mathcal{E}_{s,t}^g[\xi] + \mathcal{E}_{s,t}^g[\zeta]$.
- (v) If $\forall t > 0, x \in \mathbb{R}, \forall (y, z) \in \mathbb{R} \times \mathbb{R}, g(t, x, y, z) \geq -g(t, x, -y, -z)$, then for any $0 \leq s \leq t, \forall \xi \in \mathcal{F}_t^W, \mathcal{E}_{s,t}^g[\xi] \geq -\mathcal{E}_{s,t}^g[-\xi]$.

Proposition 1 and the above properties imply that the option price in the g -pricing framework is essentially characterized by generator g . Thus, Peng (2012) argued that the option pricing problem is equivalent to finding the generator g from actual data in the options market. Several studies, such as Yang (2006), Chen (2007) and Chen and Peng (2009) attempted to find the generator g using test methods, nonparametric statistical methods, and g -submartingale decomposition methods. This process can be viewed as an inverse problem in BSDE computations. A typical BSDE computational problem involves solving an equation using a given generator g . In contrast, we addressed the problem in which the solution of the equation (i.e., the option price) is known from actual market prices, and the task is to identify the corresponding generator g . Once the appropriate g for the market is determined, the option can be priced by numerically solving the BSDE.

In this study, we explore the g -pricing mechanism from a novel perspective. We propose a method that integrates the deep learning numerical computation of BSDEs with the identification of generator g into a single unified step. Using options market data, we performed model calibration to determine the appropriate g . The detailed algorithm is presented in Sect. 3.

The g -pricing method based on deep learning

Suppose that the price process of the underlying asset is $\{X_t\}_{t \geq 0}$, the call option price process for expiration date $T > 0$ and strike price $K > 0$ is denoted by $\{Y_t^{K,T,c}\}_{0 \leq t \leq T}$, and the corresponding put option price process is denoted as $\{Y_t^{K,T,p}\}_{0 \leq t \leq T}$. These satisfy the following FBSDEs:

$$\begin{cases} X_t = x_0 + \int_0^t bX_s \, ds + \int_0^t \sigma X_s^{\beta+1} \, dW_s, & (7a) \\ Y_t^{K,T,i} = \phi_i(X_T, K) + \int_t^T g(s, X_s, Y_s^{K,T,i}, Z_s^{K,T,i}) \, ds - \int_t^T Z_s^{K,T,i} \, dW_s, & (7b) \end{cases}$$

where i is c or p , representing the call or put options, respectively.

Compared to the general form of (1a), the forward SDE (7b) is specified as the CEV equation, as referenced in Cox (1997). When $-1 < \beta < 0$, the equation implies that the volatility of the underlying asset is a monotonically decreasing function of the asset price, which illustrates the observable leverage effect in the stock market. Specifically, volatility increases when stock prices fall and tends to decrease when stock prices rise. The initial x_0 is the known underlying price at time $t = 0$. We assumed that $b = 0$ and (σ, β) are the given parameters. Clearly, when $\beta = 0$, the CEV equation degenerates into a geometric Brownian motion process consistent with the BSM hypothesis.

We assumed that tradable European option contracts with X as the underlying asset are priced by BSDE (7b). However, the generator function, $g(\cdot)$, which serves as the pricing mechanism, is underdetermined. It is worth noting that we assume that the generator g driving pricing is the same for different $K, T, c/p$. The goal of the model is to train the g -pricing equation (7b) using deep learning techniques based on actual price data from the options market.

The underdetermined function g in Eq. (7b) leads to a distinction from the problem of numerically solving the BSDEs. In the upcoming deep learning framework, the structure of g is learned simultaneously with the solution $Y_0^{K,T,i}$ in Eq. (7b). The numerical approximation of $Y_0^{K,T,i}$ serves as an explicit output of the model.

The motivation

The forward SDE assumes that the underlying asset price follows a process with constant volatility elasticity. By combining the underlying asset with a risk-free asset, we can achieve a perfect hedge under the assumption of a complete market. Given the payoff of the option, i.e., as the terminal condition in Eq. (7b), one can determine g through replication if it is assumed that the market is complete. This is analogous to the relationship between the g -pricing mechanism and the BSM model in Sect. 2.2.

However, in this study, we do not make the additional assumption that the market is complete; rather, we want to find the g to be determined from real market data. It is well known that even when assuming the asset price process follows the CEV equation, various other factors such as trading commissions, liquidity costs, interest rates, and uncertainties in volatility still impact the fair pricing of options. This is also the reason why many other studies formulate complex stochastic equations for these factors, such as the Hull-White model (Hull and White 1987), the Heston model (Heston 1993), and the SABR model (Hagan et al. 2002). It should be noted that although these models consider complex price-influencing factors, the pricing of options is still driven by a BSDE dependent on the linear generator g , similar to the form of the BSDE corresponding to the BSM model in Sect. 2.2.

This study is motivated by an alternative perspective. Given the underlying stock price process, we attribute the complex factors influencing option prices to the

nonlinear nature of generator g of BSDE, thereby facilitating fair option pricing. In this context, the g -pricing method can be understood as a data-driven pricing instrument that can potentially supplement traditional hedging or risk-neutral pricing methods.

In addition, this study adopts the forward CEV equation to contrast it with the geometric Brownian motion assumed in the BSM model to validate the performance of the g -pricing model under different assumptions for the forward SDE (refer to the numerical results in Table 1). Beyond the SDE of CEV form, other assumptions for the forward SDE can also be incorporated into the framework of this study. Specifically, employing more accurate descriptions of the dynamics of the underlying asset prices, which are closer to real market conditions, is expected to enhance pricing performance.

The algorithm

First, we define the partition for a given expiration time, T . Because the model is required to work for different options contracts (K, T) , we introduce an isometric partition $\pi_T : 0 = t_0 < t_1 < \dots < t_{N-1} < t_N = T$ with a fixed number of steps N . For any $T > 0$, the time interval is $h_T = T/N$. The n -th time step is $t_n = nh_T, n = 0, 1, \dots, N$. The discrete scheme of the forward SDE (7a) is given by:

$$X_0^{\pi_T} = x_0, \quad (8)$$

$$X_{n+1}^{\pi_T} = X_n^{\pi_T} + bX_n^{\pi_T}h_T + \sigma \cdot (X_n^{\pi_T})^{\beta+1} \Delta W_n^{\pi_T}, \quad n = 0, 1, \dots, N-1. \quad (9)$$

For clarity, we used the simplified notation $X_n^{\pi_T}$ as $X_{t_n}^{\pi_T}$. The Brownian increment is $\Delta W_n^{\pi_T} = W_{t_{n+1}}^{\pi_T} - W_{t_n}^{\pi_T} \sim \mathcal{N}(0, h_T)$. By randomly sampling $\Delta W_n^{\pi_T}$, we simulated the trajectory of the underlying price from x_0 at time $t = 0$ to the terminal value $X_N^{\pi_T}$. Clearly, each $X_n^{\pi_T}$ computed according to the recursive formula (9) is naturally $\mathcal{F}_{t_n}^W$ -measurable, $n = 0, 1, \dots, N$.

For BSDE (7b), we adopted a backward recursive form to ensure that the output at time 0 for each stochastic path, denoted by Y_0 , precisely matches the actual market option prices. The backward recursive scheme of the BSDE (7b) is expressed as follows:

$$Y_N^{K, \pi_T, i} = \phi_i(X_N^{\pi_T}, K), \quad (10)$$

$$Y_n^{K, \pi_T, i} = Y_{n+1}^{K, \pi_T, i} + g_n^{K, \pi_T, i}h_T - Z_n^{K, \pi_T, i} \Delta W_n^{\pi_T}, \quad n = N-1, \dots, 1, 0. \quad (11)$$

As the recursive formula of the forward SDE is not affected by the BSDE, it can be computed independently up to step N to obtain $X_N^{\pi_T}$. Therefore, the terminal condition in Eq. (10) serves as the initial value for the BSDE recursive procedure. Next, it is necessary to define $g_n^{K, \pi_T, i}$ and $Z_n^{K, \pi_T, i}$ in Eq. (11) to clearly implement the backward recursive procedure.

The issue of measurability needs to be emphasized. As discussed in Sect. 2, we are seeking $\mathcal{F}_t^W$ -measurable solutions (Y_t, Z_t) . Here, Y_t represents the derivative price and Z_t contains information about the hedging strategy. The assumption of the adapted process indicates that the derivative price Y_t depends only on the information available up to the present time and, therefore, cannot acquire any future stochastic information in advance. Similarly, we require the measurability of Z_t , indicating that the hedging strategy Z_t is feasible.

The backward recursive formulation (11) defined from Y_{n+1} to Y_n inherently causes the iterative Y_n to depend on Y_{n+1} and the future stochastic information $W_{n+1} - W_n$. Consequently, the measurability of Y_n cannot be directly ensured. Therefore, this study proposes the introduction of a measurable random variable to filter future information. We introduced the adapted control terms $\{\text{net}_i^{[u]}\}_{i \in \{c,p\}}$ and $\{\text{net}_i^{[Z]}\}_{i \in \{c,p\}}$ in the form of neural networks to correct the measurability of the solution. According to the nonlinear Feynman–Kac formula, (Y_t, Z_t) can be expressed as a deterministic function of X_t . Therefore, we designed neural networks with the following structure:

$$u_n^{K,\pi_T,i} = \text{net}_i^{[u]}(t_n^{\pi_T}, X_n^{\pi_T}, K, T; \theta_i^{[u]}), \quad i \in \{c, p\}, \quad (12)$$

$$Z_n^{K,\pi_T,i} = \text{net}_i^{[Z]}(t_n^{\pi_T}, X_n^{\pi_T}, K, T; \theta_i^{[Z]}), \quad i \in \{c, p\}. \quad (13)$$

The notation $i \in \{c, p\}$ indicates that call and put options use separate networks $\text{net}_c^{[u]} / \text{net}_p^{[u]}$ and $\text{net}_c^{[Z]} / \text{net}_p^{[Z]}$, where c and p denote call and put options, respectively. Because $X_n^{\pi_T}$ for $n = 0, 1, \dots, N-1$ has already been simulated, both $u_n^{K,\pi_T,i}$ and $Z_n^{K,\pi_T,i}$ for all $n = 0, 1, \dots, N-1$ are well-defined before executing the backward recursive procedure (11). To approximate the underdetermined generator g , we introduced a neural network $\text{net}^{[g]}$ which depends on the simulated process states $X_n^{\pi_T}, u_n^{K,\pi_T,i}, Z_n^{K,\pi_T,i}$. Specifically, we defined

$$g_n^{K,\pi_T,i} = \text{net}^{[g]}(t_n^{\pi_T}, X_n^{\pi_T}, u_n^{K,\pi_T,i}, Z_n^{K,\pi_T,i}; \theta^{[g]}), \quad (14)$$

for $n = 0, 1, \dots, N-1$. Notably, unlike the networks for u and Z , network g does not need to be separated into c and p .

According to the definitions of neural networks (12)–(14), Eq. (11) can be recursively solved for $Y_0^{K,\pi_T,i}$ in the order of $n = N-1, \dots, 1, 0$. The actual options trading (or quote) price in the market is denoted as $Y_0^{*K,T,i}$. Thus, the loss function (under one trajectory of Brownian motion) is defined as follows:

$$\ell(\Theta) = |Y_0^{*K,T,i} - Y_0^{K,\pi_T,i}|^2 + \lambda \ell_y + \rho \ell_z, \quad (15)$$

where the network parameters are denoted as $\Theta = \{\{\theta_i^{[u]}\}_{i \in \{c,p\}}, \{\theta_i^{[Z]}\}_{i \in \{c,p\}}, \theta^{[g]}\}$ and the regularizations of Y and Z are defined by

$$\ell_y = \frac{1}{N} \sum_{n=0}^{N-1} |Y_n^{K,\pi_T,i} - u_n^{K,\pi_T,i}|^2, \quad (16)$$

$$\ell_z = \frac{1}{N} \sum_{n=0}^{N-1} |Z_n^{K,\pi_T,i} - \partial_x u_n^{K,\pi_T,i} \sigma \cdot (X_n^{\pi_T})^{\beta+1}|^2. \quad (17)$$

The first term of the loss function (15) calibrates the model output $Y_0^{K,\pi_T,i}$ with the market price $Y_0^{*K,T,i}$, and the last two regularization terms, ℓ_y and ℓ_z ensure that the solution satisfies Eq. (7b). Specifically, we guaranteed the measurability of Y by penalizing the distance between the measurable control u_n and the backward recursive solution Y_n . We observed that Z , as a hedging strategy, played an important role in the optimization process. Because of the variability of the backward generator g , which is represented by neural networks (14) in the optimization process, completing the optimization is more challenging than finding numerical solutions for a given BSDE with a fixed generator. The Feynman–Kac formula provides constraints that represent Z as $\partial_x u \sigma X^{\beta+1}$. Therefore, introducing the third term ℓ_z into the loss function (15) can effectively prevent the optimization process from becoming stuck at a local optimum.

To solve BSDEs based on deep learning numerical algorithms, E et al. (2017) and Han et al. (2018) designed a forward recursive scheme. Their optimization objective was to ensure that the solution of the recursive equation at the terminal time matches the target terminal condition for any given Brownian motion path. By contrast, in our proposed g -pricing method, the BSDE is approximated using a backward recursive scheme. The training objective is to ensure that the BSDE starts from the terminal condition under different stochastic paths with the option price $Y_0^{*K,T,i}$ serving as the common initial value for these paths. Furthermore, our focus lies in seeking an appropriate generator g structure within the constraints of market options data, while adjusting (Y, Z) to satisfy the BSDE associated with the nonlinear function g represented by a neural network.

Notice that Eqs. (12)–(14) contain five neural networks. Our model considers options data with different expiration dates T . In addition, we utilized a fixed number of steps N for the numerical approximation. The variable T and fixed number of steps N prevent the construction of neural networks on locked time grids. This results in an isometric N -step partition π_T that depends on the parameter T , that is, the n -th time step $t_n = nT/N$, which varies across different values of T . If $T_1 \neq T_2$, then $t_n^{\pi_{T_1}} \neq t_n^{\pi_{T_2}}$, $n = 0, 1, \dots, N$. Note that for $T_1 = 0.01$ and $T_2 = 1.5$, using a fixed step size Δt would lead to T_2 requiring 150 more iterations than T_1 . This could result in a significant increase in the number of iterations for options with longer expiration dates. Therefore, our problem is distinct from scenarios that involve solving BSDEs with a fixed terminal time T , as seen in E et al. (2017), Han et al. (2018) and Ji et al. (2020). Taking Han et al. (2018) as an example, in the context of solving the BSDEs, the terminal time T is given and fixed (for instance, assuming $T = 0.5$, $N = 10$). They introduced a network representation for Z , which, in the notation of this study, can be expressed as $Z_n^{\pi_T} = \text{net}_n^{[Z]}(X_n^{\pi_T}; \theta_n^{[Z]})$, $n = 0, 1, \dots, N-1$. Here, each network $\text{net}_n^{[Z]}$ is an independent network at time $t_n = 0.05n$, and its parameters $\theta_n^{[Z]}$ are entirely independent of the parameters $\theta_{n+1}^{[Z]}$ of the network $\text{net}_{n+1}^{[Z]}$ at time $t_{n+1} = 0.05 \times (n+1)$. However, if the terminal time is changed to $T = 1.0$, these N networks must be retrained. By contrast, we assumed that the BSDE (7b) applies to arbitrary K , c/p , and T , making it impractical to set up a network fixed at a specific time to represent Z . Therefore, we introduced the networks denoted by $Z_n^{K,\pi_T,i} = \text{net}_i^{[Z]}(t_n^{\pi_T}, X_n^{\pi_T}, K, T; \theta_i^{[Z]})$, $i \in \{c, p\}$. In other words,

apart from using two different networks for c/p , each network uses the variables t_n , T , and K as inputs. This approach allowed us to design networks that are explicitly dependent on $t_n^{\pi T}$, thereby requiring the introduction of only five neural networks that share parameters across different time steps.

The four steps of the deep learning g -pricing approach are summarized as follows:

- step 1: For a given expiration time T , the time interval is $h_T = T/N$. Randomly sample to obtain independent Brownian motion increments;
- step 2: Initialize the forward SDE with the given underlying price $X_0^{\pi T} = x_0$ and iterate to $X_{n+1}^{\pi T}$ ($n = 0, \dots, N-1$) using Eq. (9);
- step 3: For a given option (K, T, i) , define the neural network $\{\text{net}_i^{[u]}\}_{i \in \{c,p\}}$, $\{\text{net}_i^{[Z]}\}_{i \in \{c,p\}}$ and $\text{net}^{[g]}$ according to Eqs. (12)–(14). Apply backward recursion to $\{Y_n^{K,\pi T,i}\}_{0 \leq n \leq N-1}$ through Eqs. (10)–(11);
- step 4: Repeat the sampling for the Brownian motion and complete steps 1–3. Calculate the average loss according to the loss function (15), and apply the gradient method to optimize the network parameters Θ .

The complete computational pseudo-code is provided in Algorithm 1.

Algorithm 1 Deep Learning g -pricing Algorithm

Input: underlying price x_0 , maturity T , strike K , actual (call or put) option market prices $Y_0^{*K,T,i}$, time partition $h_T = T/N$, Brownian motion incremental path $\{\Delta W_n^{\pi T}\}_{0 \leq n \leq N-1}$, parameters of the forward SDE (b, σ, β) , weights of loss λ and ρ , learning rate η .

Output: $Y_0^{K,\pi T,i}(x_0)$, $i \in \{c, p\}$.

```

1:  $X_0^{\pi T} \leftarrow x_0$ 
2: for  $n = 0, 1, \dots, N-1$  do
3:    $X_{n+1}^{\pi T} \leftarrow X_n^{\pi T} + bX_n^{\pi T}h_T + \sigma \cdot (X_n^{\pi T})^{\beta+1}\Delta W_n$  ▷ forward recursion (9)
4: end for
5:  $Y_N^{K,\pi T,i} \leftarrow \phi_i(X_N^{\pi T}, K)$ 
6:  $\ell_y \leftarrow 0$ 
7:  $\ell_z \leftarrow 0$ 
8: for  $n = N-1, \dots, 1, 0$  do
9:    $u_n^{K,\pi T,i} \leftarrow \text{net}_i^{[u]}(t_n^{\pi T}, X_n^{\pi T}, K, T; \theta_i^{[u]})$ 
10:   $Z_n^{K,\pi T,i} \leftarrow \text{net}_i^{[Z]}(t_n^{\pi T}, X_n^{\pi T}, K, T; \theta_i^{[Z]})$ 
11:   $g_n^{K,\pi T,i} \leftarrow \text{net}^{[g]}(t_n^{\pi T}, X_n^{\pi T}, u_n^{K,\pi T,i}, Z_n^{K,\pi T,i}; \theta^{[g]})$ 
12:   $Y_n^{K,\pi T,i} \leftarrow Y_{n+1}^{K,\pi T,i} + g_n^{K,\pi T,i}h_T - Z_n^{K,\pi T,i}\Delta W_n^{\pi T}$  ▷ backward recursion (11)
13:   $\ell_y \leftarrow \ell_y + \frac{1}{N}|Y_n^{K,\pi T,i} - u_n^{K,\pi T,i}|^2$ 
14:   $\ell_z \leftarrow \ell_z + \frac{1}{N}|Z_n^{K,\pi T,i} - \partial_x u_n^{K,\pi T,i}\sigma \cdot (X_n^{\pi T})^{\beta+1}|^2$ 
15: end for
16:  $\ell \leftarrow |Y_0^{*K,T,i} - Y_0^{K,\pi T,i}|^2 + \lambda\ell_y + \rho\ell_z$ 
17:  $\Theta \leftarrow \Theta - \eta\nabla_{\Theta}\ell$  ▷ Neural network parameters  $\Theta = \{\theta_i^{[u]}, \theta_i^{[Z]}, \theta^{[g]}\}$  updated by gradient descent

```

Input: underlying price x_0 , maturity T , strike K , actual (call or put) option market prices $Y_0^{*K,T,i}$, time partition $h_T = T/N$, Brownian motion incremental path $\{\Delta W_n^{\pi_T}\}_{0 \leq n \leq N-1}$, parameters of the forward SDE (b, σ, β) , weights of loss λ and ρ , learning rate η .

Output: $Y_0^{K,\pi_T,i}(x_0)$, $i \in \{c, p\}$.

- 1: $X_0^{\pi_T} \leftarrow x_0$
- 2: **for** $n = 0, 1, \dots, N-1$ **do**
- 3: $X_{n+1}^{\pi_T} \leftarrow X_n^{\pi_T} + bX_n^{\pi_T}h_T + \sigma \cdot (X_n^{\pi_T})^{\beta+1}\Delta W_n$ $\triangleright$ forward recursion (9)
- 4: **end for**
- 5: $Y_N^{K,\pi_T,i} \leftarrow \phi_i(X_N^{\pi_T}, K)$
- 6: $\ell_y \leftarrow 0$
- 7: $\ell_z \leftarrow 0$
- 8: **for** $n = N-1, \dots, 1, 0$ **do**
- 9: $u_n^{K,\pi_T,i} \leftarrow \text{net}_i^{[u]}(t_n^{\pi_T}, X_n^{\pi_T}, K, T; \theta_i^{[u]})$
- 10: $Z_n^{K,\pi_T,i} \leftarrow \text{net}_i^{[Z]}(t_n^{\pi_T}, X_n^{\pi_T}, K, T; \theta_i^{[Z]})$
- 11: $g_n^{K,\pi_T,i} \leftarrow \text{net}_i^{[g]}(t_n^{\pi_T}, X_n^{\pi_T}, u_n^{K,\pi_T,i}, Z_n^{K,\pi_T,i}, \theta_i^{[g]})$
- 12: $Y_n^{K,\pi_T,i} \leftarrow Y_{n+1}^{K,\pi_T,i} + g_n^{K,\pi_T,i}h_T - Z_n^{K,\pi_T,i}\Delta W_n^{\pi_T}$ $\triangleright$ backward recursion (11)
- 13: $\ell_y \leftarrow \ell_y + \frac{1}{N}|Y_n^{K,\pi_T,i} - u_n^{K,\pi_T,i}|^2$
- 14: $\ell_z \leftarrow \ell_z + \frac{1}{N}|Z_n^{K,\pi_T,i} - \partial_x u_n^{K,\pi_T,i} \sigma \cdot (X_n^{\pi_T})^{\beta+1}|^2$
- 15: **end for**
- 16: $\ell \leftarrow |Y_0^{*K,T,i} - Y_0^{K,\pi_T,i}|^2 + \lambda\ell_y + \rho\ell_z$
- 17: $\Theta \leftarrow \Theta - \eta \nabla_{\Theta} \ell$ $\triangleright$ Neural network parameters $\Theta = \{\theta_i^{[u]}, \theta_i^{[Z]}, \theta_i^{[g]}\}$ updated by gradient descent

Empirical analysis on SPX options

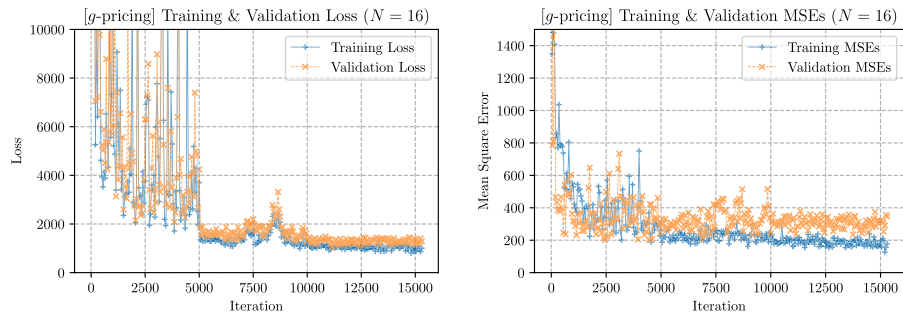
In this section, we trained the g -pricing model using the actual market prices of options and demonstrated the effectiveness of the proposed g -pricing algorithm. The model comparison was primarily based on the BSM model. Under the same underlying price assumptions, the price in the g -pricing model is closer to the market price.

SPX option dataset

We extracted historical daily frequency data of SPX options from October 1, 2020, to December 20, 2020, from the WRDS OptionMetric database. Excluding the weekly SPX options (SPXW), there are 100,521 data records with various expiration dates and strike prices. The dataset contains the following information: date (t), daily closing price of the S&P 500 index (X), strike price and expiration date (K, T) of the option contract, and maximum bid and minimum ask price of the call and put options at closing time. We used mid-price $(\text{Ask} + \text{Bid})/2$ as the option's market price. The training set contained 75,540 rows (75%) and the remaining 24,981 rows were used as the validation set (25%).

Deep neural network structures and parameter settings

For $i = c$ or p , the structures of neural networks $\text{net}_i^{[u]}$ and $\text{net}_i^{[Z]}$ are identical, each comprising 9 layers of fully connected networks. $\text{net}_i^{[u]}$ and $\text{net}_i^{[Z]}$ rely on $(t_n^{\pi_T}, X_n^{\pi_T}, K, T)$. We set the input layer structure as follows:



(a) The loss function for g -pricing during training (b) The error (MSEs) for g -pricing during training
Fig. 1 The loss and error in the training process of deep learning g -pricing models ($b = 0, \sigma = 0.2, \beta = 0$)

$$\begin{aligned}
 t_n &\mapsto i_n^t \in \mathbb{R}^{50}, \\
 X_n &\mapsto i_n^X \in \mathbb{R}^{50}, \\
 K &\mapsto i_n^K \in \mathbb{R}^{50}, \\
 T &\mapsto i_n^T \in \mathbb{R}^{50}, \\
 i_n &= [i_n^t, i_n^X, i_n^K, i_n^T] \in \mathbb{R}^{200}.
 \end{aligned}$$

The superscript π_T is omitted from the above expression. In the input layer, the notation “ $\mapsto$ ” represents a separate fully connected layer with 50 nodes, and the activation function is ReLU.¹ The input layer vectors are fed into the hidden layer.

$$i_n \mapsto h_n^1 \mapsto h_n^2 \mapsto h_n^3 \mapsto h_n^4 \in \mathbb{R}^{256}.$$

Each hidden layer contained 256 nodes and the activation function was set to ReLU. Finally, a one-dimensional output representation for u and Z was obtained using a fully connected layer with no activation function:

$$h_n^4 \mapsto o_n \in \mathbb{R}.$$

The neural network $\text{net}^{[g]}$ depends on the variables $(t_n^{\pi_T}, X_n^{\pi_T}, u_n^{K, \pi_T, i}, Z_n^{K, \pi_T, i})$; thus, the structure of the input layer is set as follows:

$$\begin{aligned}
 t_n &\mapsto i_n^t \in \mathbb{R}^{50}, \\
 X_n &\mapsto i_n^X \in \mathbb{R}^{50}, \\
 u_n &\mapsto i_n^u \in \mathbb{R}^{50}, \\
 Z_n &\mapsto i_n^Z \in \mathbb{R}^{50}, \\
 i_n &= [i_n^t, i_n^X, i_n^u, i_n^Z] \in \mathbb{R}^{200}.
 \end{aligned}$$

Superscripts K, π_T, i were omitted from the above expression. The structure of the hidden and output layers from i_n to o_n of neural network $\text{net}^{[g]}$ is identical to that of $\text{net}_i^{[u]}$ and $\text{net}_i^{[Z]}$, and will not be repeated here.

¹ A fully connected layer $x \mapsto z$ is defined as $z = \text{ReLU}(Wx + b)$ and the trainable parameters of the layer are W and b . The ReLU activation function $\text{ReLU}(x) = x^+$.

Table 1 The performance of deep learning g -pricing model versus BSM model on SPX options

Model	Parameters	Training set			Validation set		
		MSE	MAE	MAPE	MSE	MAE	MAPE
BSM	$\sigma = 0.2$	1158.23	23.01	0.2319	1568.92	25.37	0.4258
g -pricing	$\sigma = 0.2, \beta = 0$	187.88	10.11	0.1895	285.73	13.31	0.2889
g -pricing	$\sigma = 0.25, \beta = -0.05$	177.90	9.80	0.1366	265.04	12.95	0.2215
g -pricing	$\sigma = 0.15, \beta = 0.05$	209.06	10.65	0.1889	323.67	14.17	0.3255

Table 2 Deep learning g -pricing versus BSM for pricing errors with different levels of moneyness (validation set)

Model	Moneyness (%)		Call options			Put options		
	Min	Max	MSE	MAE	MAPE	MSE	MAE	MAPE
BSM ($\sigma = 0.2, \beta = 0$)	MIN	−10	1678.61	24.62	0.0229	3100.01	44.69	0.8450
	−10	−5	478.86	12.02	0.0285	2772.88	40.60	0.4812
	−5	0	698.69	15.86	0.0691	1838.43	28.15	0.2028
	0	5	1479.06	30.64	0.5427	1290.46	19.95	0.0799
	5	10	2208.95	36.92	1.1946	1154.22	16.81	0.0381
	10	MAX	2578.39	34.58	1.7668	11053.57	77.64	0.0637
g -pricing ($\sigma = 0.2, \beta = 0$)	MIN	−10	128.34	8.18	0.0117	86.06	7.13	0.1751
	−10	−5	346.01	15.86	0.0444	310.42	15.80	0.3181
	−5	0	493.32	19.81	0.1103	489.94	20.35	0.2976
	0	5	565.84	21.29	0.5687	571.46	21.96	0.1320
	5	10	467.65	18.30	0.7672	457.90	18.66	0.0568
	10	MAX	291.51	12.38	0.8979	238.97	11.98	0.0156

The number of time steps was set to $N = 16$. The loss function weight parameter $\lambda = \rho = 1$. We set the size of the mini-batch to $I = 128$ and employed the Adam optimizer to optimize the parameters. The learning rate η was initialized to 10^{-3} and reduced to one-tenth of the current learning rate after every 5000 mini-batch training iterations. The training process was terminated after 15,000 iterations. The above parameter settings, network structure settings, and training procedure were general for all experiments in this section.

Experimental results

We specified a set of forward SDE parameters: $b = 0, \sigma = 0.2, \beta = 0$. This set of parameters is consistent with the BSM model's assumptions regarding the pricing process of the underlying assets. Thus, we can intuitively compare the differences between the BSM and the g -pricing model proposed in this study.

Figure 1 shows the loss function and mean squared errors (MSEs) of the g -pricing model during training in the case of $b = 0, \sigma = 0.2, \beta = 0$. The values of MSEs represent the mean-square error between market option prices and model outputs in each mini-batch. Because the learning rate drops to one-tenth of the previous level after every 5000 mini-batch iterations, the error varies drastically in the first 5000

iterations and then smooths out. The loss functions for the training and validation sets exhibited synchronized downward trends, and there was no significant numerical difference between the two sets. These characteristics indicate that the model did not exhibit significant overfitting. The loss function is defined in Eq. (15), which consists of three parts. The first is the error between the outputs of the backward recursion and the real market prices of options. The second part is the regularization term for solving Y and the third part is the regularization term for Z . The training results of the model were aimed at achieving a balance between minimizing these three losses. We visualized the first component—the pricing error MSEs—in Fig. 1b. The MSEs for the validation set remained stable and did not increase during the latter half of the training process. We attributed this stability to the introduction of regularization constraints for Y and Z in the g -pricing model. Their presence constrains the pricing output to align with the pricing mechanism depicted by the BSDE, thereby mitigating overfitting of market data to a certain extent.

In Table 1, we compared the performance of the BSM and g -pricing models under different parameter settings. First, we analyzed the results under the parameters $\sigma = 0.2$ and $\beta = 0$, which are shown in the first two rows of Table 1. The underlying asset processes are identical in both models, and the difference in the pricing results can be attributed to the distinction between the (linear) BSDEs corresponding to BSM model and the (nonlinear) BSDEs trained from g -pricing. The MSEs, MAEs (mean absolute errors), and MAPEs (mean absolute percentage errors) of the g -pricing model are significantly smaller than those of the BSM model in both the training and validation sets, especially for the absolute error, MSEs and MAEs. Under the conditions of employing the same forward SDEs (specifically, the $\sigma = 0.2$ BSM model and the g -pricing model with $\sigma = 0.2$ and $\beta = 0$), employing the nonlinear BSDEs represented by g -pricing yields lower pricing errors than the linear BSM model. The difference in the relative errors between the two models is not significant; the reason for this is analyzed in detail in Sect. 5.

We conducted a more detailed analysis of the errors for call/put options at different levels of moneyness on the validation set, as shown in Table 2. The moneyness is denoted by $K/X - 1$, where X is the price of the underlying asset. We defined the intervals of varying degrees of at-the-money (ATM), in-the-money (ITM), and out-of-the-money (OTM) using -10% , -5% , 0% , 5% , 10% . Intervals with lower degrees of OTM/ITM were closer to 0. In the same row of the table, the higher OTM level of a call option, the higher ITM level of the corresponding put option, and vice versa. Table 2 shows that the pricing accuracies of both the BSM and g -pricing models are related to moneyness. The relative error in the pricing of ITM options is generally lower than that in the pricing of OTM options. The pricing accuracy of the OTM options determines the general performance of the model. The deep learning g -pricing model has an absolute error of 7.13 and a relative error of 0.1751 in the deep OTM [MIN, -10%] of the put options, whereas the BSM model has an absolute error of 44.69, and a relative error of 0.8450.

Regarding the performance of the pricing model, the MSEs for ATM options are higher than those for OTM options. However, this difference does not necessarily imply that the model performs worse for ATM options than for OTM options. The primary reason

for this outcome is the significant price disparity between ATM and OTM options, with OTM option prices being much smaller. Therefore, to evaluate the performance of the model in these two regions, a more meaningful measure is the relative error, specifically MAPEs. Notably, the MAPEs for the ATM options were significantly smaller than those for OTM options.

From the framework of this study, the parameters of the forward SDE and generator g of the BSDE jointly determine the accuracy of the pricing model. Table 1 shows the results for the other two sets of forward SDE parameters: $(\sigma = 0.25, \beta = -0.05)$ and $(\sigma = 0.15, \beta = 0.05)$. In Table 1, three sets of distinct forward SDE parameters were employed. When $\sigma = 0.25$ and $\beta = -0.05$, the pricing errors in both the training and validation samples are lower than those in the other two sets. $\beta < 0$ characterizes the leverage effect of the underlying volatility, which is usually observable. This suggests that there is a compelling rationale to believe that further refinement in estimating the parameters σ and β through historical stock prices would enhance the overall performance of the pricing model. The purpose of conducting the experiment with the three sets of parameters corresponding to g -pricing in Table 1 is to demonstrate this viewpoint.

Discussion

In this section, we focus on the relative error problem in option pricing. In the empirical analysis, we find that, although the deep learning g -pricing model has high absolute accuracy, the relative error is not significantly different from that of the BSM model under the same forward SDE.

The loss function (15) defines the training target as a mean-squared loss. For a deeply ITM call option, its price is close to the underlying price X_t because the strike price K is small. Then, a put option with the same strike price K is deeply OTM and the price is close to zero. Using the European call option as an example, according to the no-arbitrage principle, $\max(X_t - Ke^{-r(T-t)}, 0) \leq Y_t^{K,T,i} \leq X_t$. When $t \rightarrow T$ and $K \gg X_t$, the option is a deep OTM and $Y_t^{K,T,i} \rightarrow 0$; otherwise, when $t \rightarrow T$ and $K \rightarrow 0$, the option is a deep ATM and $Y_t^{K,T,i} \rightarrow X_t$. Given that the underlying asset price fluctuated around 3000 during the selected time period for SPX option data, it is possible that ATM option prices exceeded 2,000, while concurrently, OTM option prices could be lower than 0.1. Under mean-squared loss, ITM option prices with larger magnitudes are more thoroughly optimized. However, the relative error is also important for option pricing. Because the minimum tick size of SPX is 0.05, the error of one tick has almost no effect on ITM options, but the relative error caused by deep OTM options with a price of 0.1 is as high as 50%.

In response, we discuss a method for addressing the issue of relative error. Generally, for any $0 \leq t \leq T$, the underlying price $X_t > 0$, the option price $Y_t^{K,T,i} \geq 0$, and the terminal condition $\phi_i(X_T, K) \geq 0$. Let $x_t = \log X_t$ and $y_t^{K,T,i} = \log(Y_t^{K,T,i} + \delta)$, $z_t^{K,T,i} = \frac{Y_t^{K,T,i}}{Y_t^{K,T,i} + \delta}$, where δ is a specified constant. Consider the following FBSDEs:

$$x_t = x_0 + \int_0^t -\frac{1}{2}e^{-2x_s}\sigma^2(s, e^{x_s}) ds + \int_0^t e^{-x_s}\sigma(s, e^{x_s}) dW_s, \quad (18)$$

$$y_t^{K,T,i} = y_T^{K,T,i} + \int_t^T \tilde{g}(s, x_s, y_s^{K,T,i}, z_s^{K,T,i}) ds - \int_t^T z_s^{K,T,i} dW_s, \quad (19)$$

where $\sigma(s, x) = \sigma x^{\beta+1}$, $y_T^{K,T,i} = \log(\phi_i(X_T, K) + \delta)$. The SDE (18) can be easily obtained from Eq. (7b) based on the Itô formula, where $b = 0$. Similarly, applying the Itô formula to the BSDE (7b) yields Eq. (19), where $\tilde{g}(s, x, y, z) = g(s, e^x, e^y - \delta, ze^y)e^{-y} + \frac{1}{2}z^2$. Thus, when the function $\tilde{g}$ is recognizable, the generator g can be reproduced.

After taking the logarithm, the BSDE (19), generator $\tilde{g}$, and model loss MSEs would depend on the logarithmic prices, which show relatively minor variations in magnitude. As zeros may appear in the option price and its terminal conditions, it is necessary to introduce a constant δ when taking the logarithm. For example, we set $\delta = e^{0.05}$, where 0.05 is the minimum change in the price at which the SPX options contract trades $\epsilon = 0.05$. We checked the relationship between the logarithmic price and original option price: (i) If $Y_t^{K,T,i} \rightarrow 0$, then $y_t^{K,T,i} \rightarrow \epsilon$. (ii) When ϵ is small, $y_t^{K,T,i} = \log(Y_t^{K,T,i} + e^\epsilon)$ is of the same order as $Y_t^{K,T,i}$ near 0. The properties above indicate that $y_t^{K,T,i}$ is strictly greater than zero. It is of the same order as $Y_t^{K,T,i}$ for small values and has no large order-of-magnitude difference between OTM and ITM.

However, empirical evidence indicates that this method makes it difficult to obtain low relative errors. In contrast, this increases the absolute error. We analyzed that the reason may be that the generator $\tilde{g}$ of BSDE (19) grows quadratically concerning z , and the terminal condition $y_T^{K,T,i}$ is not bounded. Referring to Briand and Hu (2006), the rapid movement of the $\tilde{g}$ term of BSDE (19) may cause its numerical computation to become extremely unstable, affecting the convergence of the pricing error. The issue of relative error handling in deep-learning g -pricing methods remains a direction for future work.

Conclusion

This study proposes a deep-learning-based g -pricing method for options. The g -pricing mechanism proposed by Peng (2012) and its subsequent g -expectation theory introduced a dynamic approach for derivative pricing. The generator g in this theory determines the nature of the pricing. Therefore, finding a generator g that matches market pricing rules is an interesting topic. To achieve this objective, we adopted a data-driven approach. Given a forward SDE, the backward recursive formulation of the BSDE is obtained by simultaneously introducing a neural network control term (Y, Z, g) . This yields the difference between the numerical solution at $t = 0$ and the market price of real options.

We empirically validated the model's performance on SPX options. The empirical results show that the deep learning g -pricing model has lower absolute errors than the BSM model for the same forward SDE. The g -pricing mechanism has potential applications in option pricing.

In future studies, we plan to focus on three specific issues. First, when there are significant order-of-magnitude differences in the option data, the mean-square error loss function may lead to larger relative errors. Second, we explored the joint estimation of the parameters for the forward SDE and the generator of the BSDE. Finally, the robustness and computational efficiency of deep learning methods require further investigation.

Abbreviations

$\mathbb{R}^n$	n -dimensional vector space
$(\Omega, \mathcal{F}, \mathbb{P})$	The probability space
$\{X_t\}_{t \geq 0}$	The underlying asset price series
$\{Y_t\}_{t \geq 0}$	The option price series
$\{W_t\}_{t \geq 0}$	The Brownian motion series
$\{\mathcal{F}_t^W\}_{t \geq 0}$	The filtration generated by the Brownian motion
$L^2(\mathcal{F}_t^W)$	Space of $\mathbb{R}$ -valued $\mathcal{F}_t^W$ -measurable square-integrable random variables
$\mathcal{E}_{t,T}^g[\cdot]$	The g -pricing mechanism driven by the BSDE with generator g
$N(\cdot)$	The cumulative distribution function of the standard normal distribution
BSM	Black–Scholes–Merton
SDEs	Stochastic differential equations
BSDEs	Backward stochastic differential equations
FBSDEs	Forward–backward stochastic differential equations
PDEs	Partial differential equations
CEV	Constant elastic variance
SPX	The S&P 500 index
MSE	Mean square error
MAE	Mean absolute error
MAPE	Mean absolute percentage error
ITM	In-the-money
ATM	At-the-money
OTM	Out-of-the-money

Acknowledgements

We would like to thank the Managing Editor-in-Chief and the anonymous referees for their constructive comments and suggestions which helped us to improve the manuscript.

Author contributions

The authors contributed equally to this work and are listed in alphabetical order. All authors read and approved the final manuscript.

Funding

This work is supported by Taishan Scholar Project of Shandong Province of China (Grant tstp20240803), the National Key R&D Program of China (Grant No. 2023YFA1008903) and the Major Fundamental Research Project of Shandong Province of China (Grant No. ZR2023ZD33).

Availability of data and materials

All the SPX data used in this paper are collected from the WRDS OptionMetric database which can be accessed at <https://wrds-www.wharton.upenn.edu/pages/about/data-vendors/optionmetrics/>. The experimental code for this paper is available on GitHub: <https://github.com/TENGBINN/g-pricing>. Due to license agreement restrictions related to WRDS OptionMetric database, a synthetic data sample is provided on GitHub.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests

The authors declare that they have no conflict of interests.

Received: 30 August 2023 Accepted: 19 November 2024

Published online: 01 April 2025

References

- Artzner P, Delbaen F, Eber JM et al. (1999) Coherent measures of risk. *Math Financ* 9(3):203–228
- Black F, Scholes M (1973) The pricing of options and corporate liabilities. *J Polit Econ* 81(3):637–654
- Briand P, Hu Y (2006) BSDE with quadratic growth and unbounded terminal value. *Probab Theory Relat Fields* 136(4):604–618
- Chen L (2007) Numerical methods for backward stochastic differential equations, nonlinear expectation and their application in finance: g -pricing mechanism and risk measure (in Chinese). In: PhD thesis, Jinan: Shandong University
- Chen Z, Epstein L (2002) Ambiguity, risk, and asset returns in continuous time. *Econometrica* 70(4):1403–1443
- Chen L, Peng S (2009) Report on testing and finding the generating functions g of an option pricing mechanism through market data. In: *Industrial and applied mathematics in China*. World Scientific, pp 79–100
- Cohen SN, Reisinger C, Wang S (2023) Arbitrage-free neural-SDE market models. *Appl Math Financ* 30(1):1–46
- Cont R, Da Fonseca J (2002) Dynamics of implied volatility surfaces. *Quant Financ* 2(1):45–60

- Cox JC (1997) The constant elasticity of variance option pricing model. *J Portf Manag* 23(5):15–17
- Derman E, Kani I (1994) Riding on a smile. *Risk* 7(2):32–39
- Dupire B (1994) Pricing with a smile. *Risk* 7(1):18–20
- E W, Han J, Jentzen A (2017) Deep learning-based numerical methods for high-dimensional parabolic partial differential equations and backward stochastic differential equations. *Commun Math Stat* 5(4):349–380
- El Karoui N, Peng S, Quenez MC (1997) Backward stochastic differential equations in finance. *Math Financ* 7(1):1–71
- Eraker B, Johannes M, Polson N (2003) The impact of jumps in volatility and returns. *J Financ* 58(3):1269–1300
- Gatheral J (2004) A parsimonious arbitrage-free implied volatility parameterization with application to the valuation of volatility derivatives. In: *Presentation at global derivatives and risk management*, Madrid
- Gatheral J, Jacquier A (2014) Arbitrage-free SVI volatility surfaces. *Quant Financ* 14(1):59–71
- Gianin ER (2006) Risk measures via g -expectations. *Insur Math Econ* 39(1):19–34
- Guyon J, Lekeufack J (2023) Volatility is (mostly) path-dependent. *Quant Financ* 23(9):1221–1258
- Hagan PS, Kumar D, Lesniewski AS et al. (2002) Managing smile risk. *Best Wilmott* 1:249–296
- Han J, Jentzen A, E W (2018) Solving high-dimensional partial differential equations using deep learning. *Proc Nat Acad Sci* 115(34):8505–8510
- Heston SL (1993) A closed-form solution for options with stochastic volatility with applications to bond and currency options. *Rev Finan Stud* 6(2):327–343
- Hull J, White A (1987) The pricing of options on assets with stochastic volatilities. *J Financ* 42(2):281–300
- Ji S, Peng S, Peng Y et al. (2020) Three algorithms for solving high-dimensional fully-coupled FBSDs through deep learning. *IEEE Intell Syst* 35(3):71–84
- Li H, Schwab J, Antholzer S et al. (2020) NETT: solving inverse problems with deep neural networks. *Inverse Prob* 36(6):065005
- Long Z, Lu Y, Dong B (2019) PDE-Net 2.0: learning PDEs from data with a numeric-symbolic hybrid deep network. *J Comput Phys* 399:108925
- Madan DB, Carr PP, Chang EC (1998) The variance gamma process and option pricing. *Rev Financ* 2(1):79–105
- Merton RC (1973) Theory of rational option pricing. *Bell J Econ Manage Sci* 4(1):141–183
- Pardoux E, Peng S (1990) Adapted solution of a backward stochastic differential equation. *Syst Control Lett* 14(1):55–61
- Pardoux E, Peng S (1992) Backward stochastic differential equations and quasilinear parabolic partial differential equations. In: *Stochastic partial differential equations and their applications*. Springer, p 200–217
- Peng S (1997) Backward SDE and related g -expectation. In: *Pitman research notes in mathematics series*, pp 141–160
- Peng S (2012) The pricing mechanism of contingent claims and its generating function. Preprint [arXiv:1211.6525](https://arxiv.org/abs/1211.6525)
- Peng S (1991) Probabilistic interpretation for systems of quasilinear parabolic partial differential equations. *Stoch Stoch Rep* 37(1–2):61–74
- Raissi M, Perdikaris P, Karniadakis GE (2019) Physics-informed neural networks: a deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. *J Comput Phys* 378:686–707
- Yang W (2006) Applications of backward stochastic differential equations and nonlinear expectations: Riskmeasure, estimation of evaluation mechanism and option pricing (in Chinese). In: *PhD thesis*, Jinan: Shandong University

Publisher's Note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.